On-board Battery Health Prediction in Sensor Node for Sustainable IoT

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***Abstract*—This is a demo abstract based on our previously published article [1]. One of the significant shortcomings in existing data-driven battery health prediction techniques include higher computational complexity requirement to predict the battery State of Health (SOH) and the Remaining Useful Life (RUL). To mitigate this drawback, an accurate and reliable solution has been proposed for battery heath prediction in battery-operated, low-resourced IoT devices. The voltage and time-based features are extracted and the random learning algorithms with good learning performance are used for SOH and RUL prediction. The proposed solution has been accurately tested on three batteries with different charging policies.**

I. Introduction

In the modern smart world, various physical objects are capable of sensing, processing, communication and internet connectivity with the help of sensor nodes in the Internet-of-Things (IoT) [2] framework. The sensor node consists of four principal components - the sensor for data acquisition, the transceiver for data transfer to/from a local system/cloud, the micro-controller for controlling both the sensor and the transceiver, and the battery for powering the entire unit. All these components together help in continuous monitoring of various tasks in our environment. The malfunction of one or more items may indicate a fault in the associated system. Like other electronics components, the battery performance degrades with time.

The real-world deployment of IoT has led to the challenge of monitoring the health and performance of the sensor nodes themselves [3]. The rechargeable Li-ion batteries used in the IoT sensor nodes lose their charge holding capacity as they degrade with time and use [4]. Hence, in this paper, we have proposed a battery health monitoring technique, as the death of the battery leads to the complete failure of the sensor node.

II. Proposed Design for On-board Battery Health Monitoring

A unique hardware component has been introduced for Battery Health Self-monitoring in the Sensor Nodes in Sustainable IoT, as shown in Fig. 1. The battery is responsible for the power supply to all other components of the sensor node, such as sensors, controller, and transmitter. The Battery Management System (BMS) recharges the battery in every charge-discharge cycle. The voltage sensor collects the time and voltage information, which is then used to extract the Health Indicator (HI) - time interval of equal charging voltage difference (TIECVD). The extracted HIs are used to predict the battery's SOH and RUL using the ELM and RVFL algorithms. Therefore, the proposed method utilizes edge-as-a-service to analyze the HIs of battery, perform edge computing and provides the necessary services for sustainable IoT applications, along with the normal sensor node functionalities. The entire calculation is done on-board with minimal external interference, and the battery's health data is then transmitted to the cloud. The external sensors responsible for measuring the respective environmental parameters are also acquired and sent to the cloud for further processing and calculation.

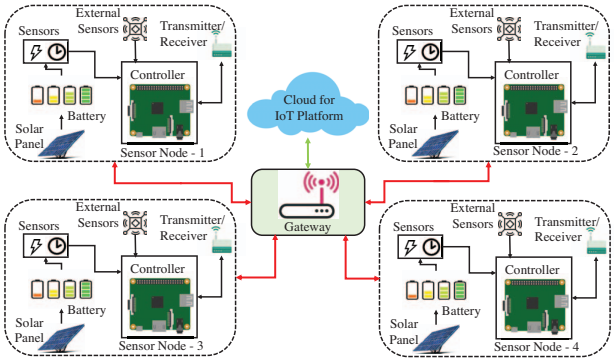

Fig. 1. Proposed architecture for on-board battery self-health monitoring in IoT sensor nodes.

III. Results and Discussion

The battery aging test dataset is drawn from experiments performed in [5], which uses Lithium Iron Phosphate (LFP) cells fabricated by A123 Systems. The nominal capacity and the nominal voltage of the rechargeable batteries under consideration are 1.1 Ah and 3.3 V, respectively. Three batteries having various charging policies and life cycles are chosen from the *Batch - 2018-04-12*.

The performance of the proposed ELM-based SOH prediction module is evaluated based on Root Mean Square

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Error (RMSE) and Mean Absolute Error (MAE). The SOH prediction results of three different batteries are summarized in Table I. For each battery, the low runtime indicates the feasibility of real-time SOH prediction using the proposed model. It is also observed that the proposed method gives high prediction accuracy irrespective of the battery charging policy.

TABLE I
SOH PREDICTION RESULTS FOR THREE DIFFERENT BATTERIES

Battery ID	Test RMSE (%)	Valid. RMSE (%)	Test MAE (%)	Valid. MAE (%)	Runtime (s)
el150800737329	0.0024	0.0063	0.0012	0.0017	0.078
el150800737270	0.0081	0.0062	0.0024	0.0044	0.0935
el150800737275	0.0053	0.0037	0.003	0.0012	0.0937

Further, the effectiveness of the proposed method is tested by executing it at different starting prediction points. Fig. 2 demonstrates the SOH prediction results for the battery ID el150800737329, when the starting points of SOH prediction are set to 60%, 70%, and 80% of the battery lifetime, respectively. This proves the efficiency of the proposed method in predicting the SOH during entire battery lifetime. It can also be observed that the accuracy increases if the prediction starts at a later point.

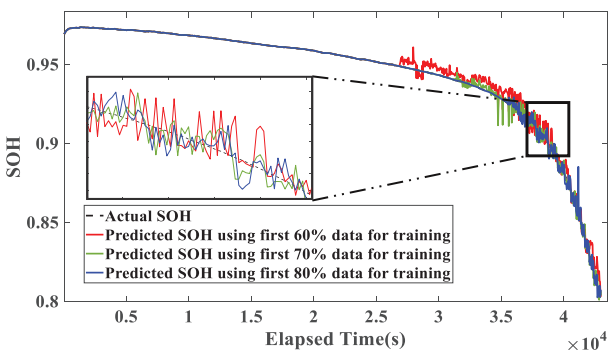

Fig. 2. Predicted SOH graph with different starting prediction points for battery ID el150800737329.

TABLE II
RUL PREDICTION RESULTS FOR THREE DIFFERENT BATTERIES

Battery ID	MSE(%)	RMSE(%)	MAE(%)
el150800737329	0.0006	0.0247	0.0163
el150800737270	0.0014	0.0384	0.0304
el150800737275	0.004	0.0216	0.0163

Similar to the SOH prediction module, the results for the RUL prediction using the RVFL network are given in terms of MSE, RMSE, and MAE. The RUL prediction results for the three previously chosen batteries are shown in the Table II. The lower values of the error metrics prove that the proposed model can accurately predict the RUL of the battery so that it can be changed timely without any loss of monitoring data. The closeness of the actual and predicted RUL for the battery ID el150800737329 is demonstrated in Fig. 3.

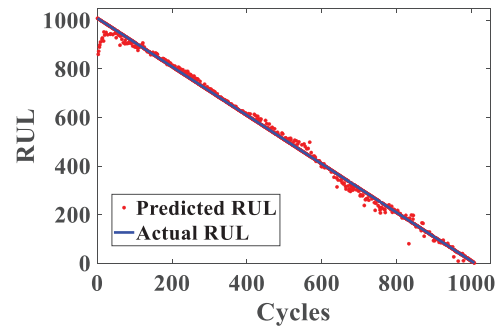

Fig. 3. Closeness of actual RUL with predicted RUL for battery ID el150800737329.

IV. Conclusion

The sudden failure of IoT sensor node due the death of the rechargeable battery can lead to significant loss of monitoring information in vital applications. Hence, a unique battery self-health monitoring architecture has been proposed in this paper to ensure uninterrupted power supply and provide ample time to the users to replace the degraded batteries in the sensor nodes. The proposed method can predict battery's SOH and RUL with 0.0024% and 0.0247% RMSE, respectively. The model is light-weight and efficient enough to be implemented in low-powered IoT devices.

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